

TEST DRIVING AUTONOMOUS VEHICLES: Smart Becoming Smarter



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Vehicles, system components and machines are becoming increasingly autonomous and network with each other via wireless technologies. Touch-screen based controls, remote diagnostics, vehicle status and control are being implemented using controller area network bus for interconnects. However, with the best of vehicles, all issues cannot be solved in the vehicle itself. Puneet Gupta, chief technology officer, Brillio Technologies, categorises the testing of modules into a couple of categories. He says, "First, we have the immediate threats that need to be handled at the user end. Then, there are some problems that can be solved in the Cloud."

Immediate threats cause physical harm

Electronic stability control (ESC) is the involvement of electronics in braking systems. Such systems help drivers avoid crashes by reducing the occurrence of skidding or losing control. Mehul Garg, embedded design engineer, Padmini VNA Mechatronics Pvt Ltd, explains, "On detecting a loss of control, a number of intelligent sensors automatically apply brakes to the relevant wheel." ESC becomes active when a driver loses control of the car. It uses computer-controlled technology to apply individual brakes and help bring the car

safely back on track, without the danger of fishtailing.

Under-steering and over-steering may be fun while playing games, but these can be disastrous in real life. Garg highlights the reasons to be wary of over-steering. Over-steering, for example, on slippery roads, could result in fishtailing and cause serious injury to multiple vehicles. These kinds of systems cannot be left to be processed in the Cloud. Immediate action has to be taken, that would require a support system after sensing a discourse.

Besides the dangers that require immediate attention, there are several threats over the long term. Infotainment, for example, might not allow a Bluetooth connection due to some issues. This can be diagnosed over the Cloud, resulting in in-vehicle processing focused on saving lives. So once we know the threats, what are the areas to test for?

Areas for focus in testing

Even with the most basic of vehicles, acceleration and deceleration form the crux of driving. These systems need to be tested properly with as low a chance of error as possible. Then, the vehicle should be checked to see if it follows the rules and guidelines of proper driving including acknowledging traffic lights and lane-driving. Any autonomous system at its heart employs a battery for operation. Nisarg Nirmalkumar, senior software developer, RA Consulting GmbH, adds, "The battery needs to be tested as well."

Connected devices prone to hacks.

Wireless connectivity features like Wi-Fi, Bluetooth and 3G or LTE present possibilities of remote attacks. Keyless entry and ignition do not require the attacker to have physical access to the vehicle. The number of wireless standards available has created the need for similar equipment to generate signals for testing those

Automotive
infotainment test
system for testing four
infotainment modules
simultaneously

